

Physically Interpretable World Models for Partially Observable Driving: Encoding and Predicting the Road Environment under Weak Supervision

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Abstract

Learning predictive models from high-dimensional sensory observations is fundamental for safe autonomous cyber-physical systems, yet the latent representations of standard world models lack physical interpretability. Our prior conference work introduced the Physically Interpretable World Model (PIWM), which aligns latent representations with physical quantities and constrains their temporal evolution through partially known dynamics, trained under distribution-based weak supervision derived from realistic sensing pipelines. That formulation implicitly assumes that the system’s physical state provides a complete Markov description of the task, a condition that holds for fully observable benchmarks such as CartPole and Lunar Lander but fails for realistic driving. Two vehicles with identical position, heading, and speed can follow divergent futures when the road geometry ahead differs, because a standard vehicle state encodes nothing about the lane or the vehicle’s relationship to it. Driving is therefore *partially observable*: the world model must reason not only about the vehicle but about the external road environment, which is neither part of the vehicle state nor fully visible from an onboard camera.

This article extends PIWM to partially observable driving along three axes. First, we move beyond the fully observable benchmarks to two new partially observable case studies: a simulated top-down *CarRacing* environment and a *real, physical DonkeyCar* platform driven on a laboratory track. Second, we introduce a physically interpretable *encoding* of the road environment: the interpretable latent is factorized into a vehicle-relative component and a road-context component (lateral deviation, vehicle–road heading error, and road curvature) that is inferred from images even when the road ahead is only partially visible. Third, we introduce a physically interpretable *prediction* of the road environment: a split-dynamics model couples a learnable

bicycle kinematic model for the vehicle with a structured, geometry-driven transition that forecasts how the road context evolves. On the real DonkeyCar, ground-truth road context is available only at training time through external track localization and is withheld at test time, so the model must recover and roll out road context from onboard images alone. Across the original benchmarks, simulated CarRacing, and the physical DonkeyCar, PIWM with factorized state and split dynamics substantially outperforms data-driven and physics-informed baselines at long prediction horizons, demonstrating that physically grounded driving world models must both encode and predict the external road environment, not merely the vehicle state.

Keywords: physically interpretable world models, partial observability, autonomous driving, road-context encoding, road prediction, bicycle dynamics, weak supervision, trajectory prediction

1. Introduction

Accurate and robust trajectory prediction from high-dimensional sensor data is a foundational challenge for autonomous cyber-physical systems (CPS). Modern *world models* [1] address this challenge by encoding observations into compact latent representations and predicting their temporal evolution, a paradigm that has proven effective for long-horizon prediction, planning, and control [2, 3, 4]. Despite strong empirical performance, the latent states of these models typically lack a clear connection to the underlying physical state of the system. In safety-critical settings such as autonomous driving and robotics, this opacity limits trustworthiness: without interpretable states, it is difficult to detect unsafe future configurations, diagnose prediction failures, or apply formal safety-monitoring techniques [5, 6, 7].

Our prior conference work introduced the Physically Interpretable World Model (PIWM) [8], a framework that aligns latent representations with physical quantities and constrains their temporal evolution through a partially known dynamics model. PIWM defines physical interpretability through two complementary properties: the latent state must correspond to meaningful physical variables (state semantics), and its temporal evolution must follow physically consistent dynamics (temporal correctness). A key feature of PIWM is *distribution-based weak supervision*: rather than requiring precise ground-truth state labels, the method accepts state estimates in the form of sample distributions, as naturally produced by real-world sensing pipelines

such as GPS or radar. Experiments on CartPole, Lunar Lander, and DonkeyCar demonstrated that PIWM recovers accurate physical parameters and achieves long-horizon prediction accuracy well beyond data-driven baselines. A systematic comparison of intrinsic versus extrinsic architectures and continuous versus discrete latent spaces showed that an extrinsic architecture with a quantized (VQ-based) latent yields the most robust and physically grounded representations.

The challenge: driving is partially observable

Applying PIWM to realistic driving exposes a fundamental limitation of the original formulation, and motivates the extensions in this article. On CartPole and Lunar Lander, the system state completely specifies what will happen next: the dynamics are fully observable and Markov, so a world model that tracks the vehicle (or pole, or lander) state captures everything relevant to the future. Driving is different. Consider two vehicles with identical position, heading, and speed. If one is approaching a tight curve while the other is on a straight segment, their future trajectories diverge substantially even under the same control inputs. The vehicle’s physical state $x_t = (x, y, \psi, v)$ does not capture this difference. The missing information is the *road context*: the shape of the lane ahead (its curvature) and the vehicle’s lateral and angular relationship to it, none of which is part of any standard vehicle-state representation. As a result, the vehicle state alone is *non-Markov* with respect to future motion: two identical states lead to different futures depending on the unobserved road geometry. Applying the original PIWM formulation to such tasks implicitly treats the system as fully observable when it is not, causing the dynamics model to learn an underdetermined, ill-posed mapping.

This challenge has two distinct facets, and overcoming each requires a separate mechanism. *(i) The road environment must be encoded.* The relevant road context is not given as a clean state vector; it is latent in the pixels and, for a forward-facing camera, only partially visible at any instant. A physically interpretable driving world model must therefore *infer* the geometry of the surrounding road from raw images, and represent it in physically meaningful terms rather than as opaque features. *(ii) The road environment must be predicted.* Long-horizon rollout requires anticipating how the road context itself evolves as the vehicle advances: the lane that is far ahead now becomes the lane the vehicle is on a few steps later. A world model that predicts only the vehicle state, treating the road as static or as noise, accumulates error rapidly because it cannot anticipate the geometry it is about

to enter. Predicting the road, like encoding it, should be physically interpretable so that the forecasted environment can be inspected, trusted, and monitored.

These two facets, encoding and predicting the external environment, are precisely what distinguishes a driving world model from the vehicle-centric formulation of the conference paper, which modeled only the state of the controlled body. They are also why partial observability cannot be papered over by simply adding capacity to a black-box predictor: the missing variable is structured, physical, and causally upstream of the vehicle’s future, so it deserves an explicit, physically grounded treatment.

Our approach

We resolve partial observability by reformulating the interpretable state as a structured composition of two complementary components. The *vehicle-relative state* z^v captures the ego vehicle’s displacement and heading change expressed in a local reference frame anchored at the beginning of each prediction window; this representation removes dependence on global coordinates and improves transferability across trajectories. The *road-context state* z^r captures task-relevant environmental structure: the cross-track error (lateral deviation from the lane center), the vehicle–road heading error, and the local road curvature. Together, (z^v, z^r) form a Markov description of the driving task, restoring the validity of the predictive world-model framework. The vehicle-relative component is governed by a learnable bicycle kinematic model, whose parameters (e.g., the effective wheelbase) are unknown and recovered during training. The road-context component is governed by a structured, geometry-driven transition that reflects the slow, conservative evolution of lane variables. This split-dynamics formulation provides appropriate inductive biases for each component and decouples the faster-varying vehicle motion from the slower-varying road context.

We validate the extended formulation in two new partially observable driving environments. In **CarRacing**, a top-down simulated racing environment, the road geometry is present in the image but absent from the vehicle state; the model must learn to infer and exploit road-context information from raw pixels. On the physical **DonkeyCar** platform, a real RC-scale car observes its environment through a front-facing camera and drives on a laboratory track. During training, an external localization system provides ground-truth road context as a privileged signal; at test time this signal is withheld, and the model must infer and roll out road context solely from

the onboard front-facing image stream. This setup tests whether the factorized representation learned under privileged supervision transfers to the more challenging unaided real-world deployment setting. Across both environments and at prediction horizons of up to 100 steps, PIWM with factorized state and split dynamics substantially outperforms data-driven baselines and physics-informed alternatives.

Contributions

This journal article substantially extends the conference version [8] along three axes, which constitute its three contributions.

1. **Two partially observable driving case studies.** We move PIWM beyond the fully observable benchmarks of the conference paper to two new driving environments that are genuinely partially observable: a simulated top-down *CarRacing* track and a *real, physical DonkeyCar* platform driven on a laboratory track under onboard-camera observation. We formally characterize partial observability in driving, showing that the vehicle state is non-Markov with respect to future trajectory, and use these two case studies to demonstrate the practical consequences and the benefit of the proposed treatment.
2. **Physically interpretable encoding of the road environment.** Unlike the conference formulation, which modeled only the controlled body’s state, we model the *external environment*. We introduce a factorization of the interpretable latent into a vehicle-relative component z^v and a road-context component z^r , where z^r comprises physically meaningful descriptors of the road (cross-track error, vehicle–road heading error, and curvature). We show how to infer z^r from images using temporal context, including the forward-camera case where the road ahead is only partially visible, and how privileged supervision available only at training time can be used to learn this encoding.
3. **Physically interpretable prediction of the road environment.** We introduce a split-dynamics model that not only rolls out the vehicle with a learnable bicycle kinematic model, but also *forecasts the road context* with a structured, geometry-driven transition. The predicted road context is itself physically interpretable, so the forecasted environment, not just the forecasted vehicle, can be inspected and monitored. We show that predicting the road, rather than treating it as static, is what enables stable long-horizon prediction under partial observability.

Throughout, we precede the technical development of each extension with the specific challenge and motivation that makes it necessary, in keeping with the two facets of partial observability identified above.

The remainder of this article is organized as follows. Section 2 surveys related work. Section 3 reviews background on autonomous CPS, world models, and physical interpretability. Section 4 formalizes the partial observability problem in driving and introduces the factorized state. Section 5 presents the PIWM architecture, the road-context encoder (Section 5.2), and the road-context predictor (Section 5.3). Section 6 describes the experimental setup, and Section 7 reports results on the benchmarks, CarRacing, and the real DonkeyCar. Section 8 discusses implications and limitations, and Section 9 concludes.

2. Related Work

We organize the discussion into five threads relevant to physically interpretable world modeling under partial observability: trajectory prediction and safe forecasting (Section 2.1), representation learning for dynamical systems (Section 2.2), world models (Section 2.3), physics-informed and interpretable learning (Section 2.4), and partial observability, privileged learning, and road structure (Section 2.5). We close each thread by positioning PIWM against it.

2.1. *Trajectory Prediction and Safe Forecasting*

Predicting future motion is central to safe planning and control [9]. Classical predictors propagate a low-dimensional state with hand-specified models: constant-velocity and Kalman-filter formulations for vehicle motion [10], and Hamilton–Jacobi reachability for sets of reachable states with formal guarantees [11, 12, 13]. These methods are transparent and certifiable, but they require an explicit dynamical model and a low-dimensional state, and they scale poorly to high-dimensional camera observations.

Learning-based predictors instead forecast directly from data. Social-pooling and generative sequence models capture multi-agent interaction [14, 15, 16], while graph- and attention-based models such as Trajectron++ achieve strong forecasting by conditioning on scene graphs and high-definition maps [17]. Hybrid schemes embed classical filters inside learned models, as

in learned Kalman filtering [18]. These models, however, depend on hand-crafted scene representations or HD-map annotations that are unavailable in many settings [19, 20], and their internal states carry no physical meaning.

A parallel line equips predictors with reliability guarantees without altering the predictor itself: conformal prediction and calibrated forecasting bound error or coverage [21, 22, 23, 24]; statistical model checking and predictive monitoring estimate the probability of property violations [25, 26, 27, 28]; and runtime and misbehavior monitors flag unsafe behavior online [29, 30, 31, 32]. These techniques are complementary to ours: they wrap a predictor with guarantees but do not make its latent state physically meaningful. Compared with this body of work, PIWM learns interpretable representations directly from raw images under weak supervision, without handcrafted scene representations or test-time state labels, so that the forecast itself, both the vehicle and the road around it, can be inspected.

2.2. Representation Learning for Dynamical Systems

Compact latent representations underpin learning from high-dimensional observations, from classical autoencoders [33] to variational autoencoders (VAEs) [34], but standard objectives impose no physically meaningful structure. A large literature seeks more structured latents by encouraging disentanglement [35, 36, 37], causal structure [38, 39], or object-centric and slot decompositions [40]. Discrete VQ-VAEs regularize the latent through a learned codebook [41, 42]; PIWM builds on this quantization but, unlike it, ties codebook components to physical values. These methods improve organization or modularity, yet none guarantees correspondence to measurable physical quantities.

Closer to physical grounding, several models steer latents toward physics. GOKU-Net restricts latent variables to plausible physical ranges without binding them to specific quantities [43]; SPARTAN produces semantically structured outputs but incorporates neither known dynamics nor action-conditioned prediction [44]; Deep Variational Bayes Filters add latent state-space dynamics to VAEs [45] but, absent physical supervision, do not recover interpretable variables; and continuous-time models such as Neural ODEs learn smooth but uninterpretable latent dynamics [46]. Representation learning specialized for pixel-based control improves downstream performance [47] while again leaving the latent opaque. Compared with these approaches, PIWM explicitly aligns latent dimensions with named physical quantities of

both the vehicle and the surrounding road, and constrains their evolution with known physics under only weak, distribution-based supervision.

2.3. *World Models*

World models learn latent dynamics for imagination-based planning and control. The Dreamer and PlaNet line learns recurrent latent state-space models from pixels [48, 49, 3]; transformer- and masked-prediction world models improve sample efficiency and long-horizon fidelity [2, 4, 50]; and a range of variants explore alternative latent factorizations [1, 51, 52]. These models excel empirically but learn latent spaces with no clear physical meaning, which limits inspection and safety monitoring.

For driving, recent world models predict rich structured futures such as 3D occupancy grids or rendered scenes [53, 54, 55, 56], improving forecasting realism but still without aligning the internal state to physical variables, and at substantial annotation and compute cost. Neuro-symbolic world models improve generalization but require predefined symbolic inputs unavailable from raw sensors [57, 58]. The closest prior work, Vid2Param, recovers physical parameters from video [59] but requires full supervision and provides no explicit world model for long-horizon rollout. Compared with these approaches, PIWM recovers physical state under weak supervision and rolls it forward through interpretable dynamics, and, new in this article, it does so for the external road environment, not only the controlled body.

2.4. *Physics-Informed and Interpretable Learning*

A complementary thread injects physics directly into learning. Sparse identification of nonlinear dynamics and differentiable-physics or end-to-end system identification recover governing equations or parameters [60, 61, 62], while physics-informed neural networks embed differential constraints in the loss [63, 64]. Within world modeling, physical priors have been added as bounds on states and actions [65, 66], physics-aware losses [67], kinematics-inspired layers [68], and Taylor-monomial-structured latent dynamics (Phy-Taylor) [69].

Most of these methods assume access to the underlying low-dimensional state and are not designed for raw visual inputs, or they enforce physical plausibility without yielding a state that names specific quantities. PIWM differs in two ways aligned with this article’s contributions: it extracts the physical state from images under weak supervision, so that physics can be applied without ground-truth labels, and it applies a structured, learnable

physical model, a bicycle model for the vehicle and a geometry-driven transition for the road, to roll the interpretable state forward. Compared with physics-informed predictors that model only the controlled body, PIWM applies interpretable physical structure to the external environment as well.

2.5. *Partial Observability, Privileged Learning, and Road Structure*

Partial observability is a long-standing challenge in robotics and reinforcement learning, classically formalized through partially observable Markov decision processes and belief-state planning [70, 71]. Model-free agents commonly cope with it by adding memory through recurrent or temporal architectures: recurrent policy gradients [72], gated recurrent networks [73], convolutional LSTMs for spatiotemporal inputs [74], and temporal convolutional networks [75]. These approaches recover an opaque belief or memory state; they do not produce a physically interpretable estimate of the missing environmental variables, which is exactly what our road-context state provides.

When information needed for learning is available only at training time, privileged-information and teacher–student learning supervise a model that must operate without it at deployment [76, 77]. This is the mechanism we use on the real DonkeyCar: ground-truth road context from training-time track localization teaches the encoder what to extract, while the deployed car observes only its front camera and must reconstruct that context from images.

In driving specifically, a substantial literature encodes lane and road structure in learned representations [17, 19], but typically requires explicit maps or HD annotations; and physics-based predictors apply kinematic or dynamic bicycle models to low-dimensional state vectors [78] rather than to learned image representations. Learning-based control has also been demonstrated on RC-scale physical cars, including imitation and reinforcement learning on platforms akin to the DonkeyCar [79, 80], but without an interpretable predictive world model. Compared with all of the above, our work is, to our knowledge, the first to integrate a factorized vehicle–road representation with a physics-informed world model trained from images under weak supervision, and to both *encode* and *predict* the road environment in physically interpretable terms.

3. Preliminaries

3.1. Autonomous CPS and World Models

Definition 1 (Autonomous CPS). An autonomous CPS $s = (X, I, Y, A, \phi_\theta, g, h)$ models the evolution of an observation-based control system, where X is the physical state space, $I \subset X$ the set of initial states, Y the observation space, A the action space, $\phi_\theta : X \times A \times \Theta \rightarrow X$ the dynamics with unknown parameters $\theta \in \Theta$, $g : X \rightarrow Y$ the observation function, and $h : Y \rightarrow A$ the controller.

The true state evolves as $x_{t+1} = \phi(x_t, a_t, \theta^*)$, where θ^* are the unknown true parameters. The controller selects actions from observations: $a_t = h(y_t) = h(g(x_t))$. A world model serves as a learned surrogate for forecasting future observations.

Definition 2 (World Model). A world model $\mathcal{W} = (\mathcal{E}, f, \mathcal{D})$ predicts the future evolution of observations in an autonomous CPS s , where $\mathcal{E} : Y \rightarrow Z$ encodes observations into latent representations, $f : Z \times A \rightarrow Z$ predicts future latents, and $\mathcal{D} : Z \rightarrow Y$ reconstructs predicted observations.

Conventional world models are evaluated by predictive accuracy in image space (e.g., MSE or SSIM). However, low image error does not imply that the underlying predicted state is physically meaningful: similar-looking frames can correspond to very different physical configurations. For safety monitoring and planning, what matters is not only visual fidelity but whether the latent state can be interpreted as a physical quantity and evolved according to physically plausible dynamics.

3.2. Physical Interpretability

Following the conference paper [8], we define *physical interpretability* via two complementary properties:

1. the latent representation z^* corresponds to meaningful physical variables (state semantics), and
2. its temporal evolution follows physically consistent dynamics (temporal correctness).

Property 1 requires alignment between the latent and the physical state, which PIWM enforces through weak supervision. Property 2 requires that multi-step rollouts of the latent state obey the known equations of motion, which PIWM enforces through the structured dynamics model.

Definition 3 (Interpretable Representation Learning Problem). *Given a dataset $\mathcal{S} = \{(y_{1:M}, a_{1:M}, \Xi_{1:M})\}_{1:N}$, where $\Xi_t = \{\xi_t^{(l)}\}_{l=1}^L$ are samples drawn from an unknown supervisory distribution $p(x_t)$, and a controller h , learn a latent encoder $\mathcal{E}$ such that the mean squared error between the physical state x_t and the encoded latent $z_t^* = \mathcal{E}(y_t)$ is minimized:*

$$\min_{\mathcal{E}} \mathbb{E}[\|x_t - \mathcal{E}(y_t)\|_2^2]. \quad (1)$$

The supervisory samples Ξ_t are drawn from an unknown distribution $p(x_t)$ with no assumed parametric form; we only require that the distribution is *mean-proximal*, meaning its mean lies within a relative interval of width δ around the true value (see Section 6.2 for the precise noise model). This is a strictly weaker assumption than standard supervised learning with precise labels, and represents the uncertainty naturally arising from real-world sensing.

4. Partial Observability in Driving

This section formalizes the challenge introduced in Section 1: in driving, the vehicle state is not a sufficient statistic for the future. The formalization motivates the factorized state that the rest of the article encodes (Section 5.2) and predicts (Section 5.3).

4.1. Why Vehicle State Is Insufficient for Driving

The PIWM conference paper formulated physical interpretability under the implicit assumption that the vehicle’s physical state x_t provides a complete, Markov description of the system: knowing x_t and action a_t determines x_{t+1} . This assumption holds for CartPole, Lunar Lander, and other low-dimensional benchmark tasks, where the system state encodes all task-relevant information. It fails for driving tasks.

In a driving task, the vehicle state $x_t = (x, y, \psi, v)$ (position, heading, speed) does not encode the geometry of the road ahead. The actual lane-following dynamics of the vehicle depend on this geometry:

$$x_{t+1} = \phi(x_t, r_t, a_t, \theta^*), \quad (2)$$

where r_t denotes the *road-context state*, which includes quantities such as the lateral deviation from the lane center, the vehicle’s heading relative to the

road tangent, and the local curvature. The road-context component r_t is not directly observed by the onboard camera and is not part of any standard vehicle-state representation.

Definition 4 (Partial Observability in Driving). *A driving task is partially observable with respect to world modeling if the vehicle state x_t alone is not sufficient to determine future motion; equivalently, if there exist two system configurations $(x_t, r_t^{(1)})$ and $(x_t, r_t^{(2)})$ with $r_t^{(1)} \neq r_t^{(2)}$ such that, under the same controller h , the resulting trajectories satisfy $x_{t+1:T}^{(1)} \neq x_{t+1:T}^{(2)}$.*

Figure 1 illustrates this definition. Two vehicles at identical positions on the track, with the same speed and heading, enter different road segments (one straight, one curved). Despite identical vehicle states and identical control actions, the predicted future trajectories diverge, because the curvature of the upcoming road introduces a lateral displacement that the vehicle-state-only dynamics cannot anticipate. Standard PIWM, which learns a single interpretable state aligned with x_t , cannot represent this difference. Two consequences follow, corresponding to the two facets of the challenge from Section 1: the model must *encode* r_t from observations (Section 5.2), and it must *predict* how r_t evolves over the horizon (Section 5.3).

4.2. Factorized Interpretable State

To resolve the partial observability problem, we augment the interpretable state with road-context information and express the vehicle component in a translation- and rotation-invariant reference frame.

Vehicle-relative state. Global coordinates (x, y) introduce unnecessary dependence on the starting position of each trajectory, complicating generalization across scenes. Instead, we express vehicle motion relative to the initial timestep of each prediction window. For a window beginning at time t_0 , the vehicle-relative state is:

$$z_t^v = \begin{bmatrix} \cos(\psi_{t_0}) (x_t - x_{t_0}) + \sin(\psi_{t_0}) (y_t - y_{t_0}) \\ -\sin(\psi_{t_0}) (x_t - x_{t_0}) + \cos(\psi_{t_0}) (y_t - y_{t_0}) \\ \psi_t - \psi_{t_0} \end{bmatrix} = \begin{bmatrix} x_{\text{rel}} \\ y_{\text{rel}} \\ \psi_{\text{rel}} \end{bmatrix}. \quad (3)$$

This transformation rotates the global displacement into the local frame of the vehicle at t_0 , so that x_{rel} is the longitudinal displacement, y_{rel} the lateral displacement, and ψ_{rel} the cumulative heading change.

Road-context state. The road-context component z_t^r captures the vehicle’s relationship to the road geometry:

$$z_t^r = \begin{bmatrix} e_{\text{CTE}} \\ e_\psi \\ \kappa \end{bmatrix}, \quad (4)$$

where e_{CTE} is the cross-track error (signed lateral distance from the lane center), e_ψ is the heading error (difference between the vehicle’s heading and the road tangent at its current position), and κ is the local road curvature. Together, z_t^r encodes the information needed to determine how the vehicle will interact with the road geometry over the coming steps.

Factorized state. The complete *factorized interpretable state* is:

$$z_t^* = [z_t^v, z_t^r] \in \mathbb{R}^6. \quad (5)$$

The pair (z_t^v, z_t^r) satisfies the Markov property under the true augmented dynamics (2): knowing the current factorized state and the action is sufficient to determine the next state, up to the unknown parameters θ^* .

4.3. Extended Learning Problem

The learning problem for partially observable driving extends the original PIWM formulation as follows.

Problem 1 (Interpretable State Learning under Partial Observability). *Given a dataset $\mathcal{S} = \{(y_{1:M}, a_{1:M}, \Xi_{1:M}^v, \Xi_{1:M}^r)\}_{1:N}$ where Ξ_t^v and Ξ_t^r are supervisory samples for the vehicle-relative and road-context components respectively, and a controller h , learn a factorized encoder such that $z_t^v = \mathcal{E}^v(y_{1:t})$ approximates the vehicle-relative state and $z_t^r = \mathcal{E}^r(y_{1:t})$ approximates the road-context state, while the pair (z_t^v, z_t^r) supports accurate future prediction under the task dynamics.*

Two aspects distinguish this from the conference formulation, and they map directly onto the two extensions developed next. First, the encoder is conditioned on a *history* $y_{1:t}$ rather than a single frame: road context cannot be inferred from a single image (particularly in the front-camera setting) and requires temporal context (Section 5.2). Second, the supervision is split: Ξ_t^v can often be computed directly from the vehicle state (available in simulation

or via external localization), while Ξ_t^r requires knowledge of the road geometry. In the real DonkeyCar experiments, road-context supervision is available only at training time and withheld at test time, so the encoder must learn to recover road context from images alone (Section 5.2), and the predictor must roll it forward (Section 5.3).

5. PIWM Architecture

The PIWM architecture retains the core structure from the conference version [8], comprising a physically interpretable autoencoder and a learnable dynamics model, and extends it with the factorized state along two axes: a road-context *encoder* (Section 5.2) and a road-context *predictor* (Section 5.3). A high-level overview is shown in Figure 2.

5.1. Learning Interpretable Representations

The autoencoder maps high-dimensional observations to a physically interpretable latent state aligned with the true physical state. We follow the conference paper in considering two architectural approaches.

Intrinsic encoding.. The encoder $\mathcal{E} : Y \rightarrow Z^*$ maps an observation directly to the interpretable latent, partitioned as $z^* = [z_p^*, z_v^*]$, where z_p^* is the physically interpretable component and z_v^* captures the residual visual information needed for reconstruction. For continuous latent spaces, the objective follows a β -VAE formulation with interpretability alignment applied to z_p^* :

$$\mathcal{L}_{\text{intrinsic-cont}} = \mathcal{L}_{\text{recon}}(y, \hat{y}) + \lambda_{\text{interp}} \mathcal{L}_{\text{interp}}(z_p^*, \Xi) + \beta D_{\text{KL}}(q(z_v^*|y) \parallel \mathcal{N}(0, I)). \quad (6)$$

For discrete latent spaces, the codebook is partitioned so that each entry $\mathbf{e}_k = [\mathbf{e}_k^p, \mathbf{e}_k^v]$ has a fixed physical component and a learnable visual component. The interpretability loss is applied to the averaged physical part of the selected codebook entries.

Extrinsic encoding.. The extrinsic approach decouples visual perception from physical interpretation via a two-stage process. In the first stage, a vision autoencoder $(\mathcal{E}_v, \mathcal{D}_v)$ is trained on image reconstruction only, producing an intermediate latent $z = \mathcal{E}_v(y)$. For continuous variants this is a β -VAE; for discrete variants a VQ-VAE with a 512-entry codebook. In the second stage,

the vision encoder is frozen and a physical encoder $\mathcal{E}_p$ is trained to map z to the interpretable state $z^* = \mathcal{E}_p(z)$, optimizing:

$$\mathcal{L}_{\text{physical}} = \lambda_{\text{interp}} \mathcal{L}_{\text{interp}}(z^*, \Xi) + \lambda_{\text{latent}} \mathcal{L}_{\text{recon}}(z, \mathcal{D}_p(z^*)). \quad (7)$$

The physical encoder $\mathcal{E}_p$ is implemented as a Transformer over the temporal latent sequence (Section 5.2), with mean pooling and linear projection to the factorized state $z^* = [z^v, z^r]$.

The conference paper’s systematic comparison showed that the extrinsic approach with a discrete (VQ) latent consistently yields the strongest physical grounding and prediction accuracy; we adopt this configuration as the primary model for the driving experiments.

Interpretability loss.. For all variants, alignment between the encoded state and the physical variables is enforced through the mean squared error to the empirical mean of the supervisory samples:

$$\mathcal{L}_{\text{interp}}(z^*, \Xi) = \|z^* - \hat{\mu}_{\Xi}\|_2^2, \quad \hat{\mu}_{\Xi} = \frac{1}{L} \sum_{l=1}^L \xi^{(l)}. \quad (8)$$

When applied to the factorized state, this loss is evaluated separately on the vehicle-relative and road-context components with supervision from Ξ^v and Ξ^r :

$$\mathcal{L}_{\text{interp}}^{\text{drive}} = \lambda_v \mathcal{L}_{\text{interp}}(z^v, \Xi^v) + \lambda_r \mathcal{L}_{\text{interp}}(z^r, \Xi^r). \quad (9)$$

The separate weighting of the road-context term λ_r reflects its central role: it is the term that forces the encoder to commit capacity to the external environment rather than only to the vehicle.

5.2. Encoding the Road Environment

Challenge and motivation.. The conference model encoded only the controlled body’s state, which for the benchmarks was directly inferable from a single frame. Road context is different in two ways. First, it is a property of the *environment*, not the agent, so it must be read off the scene rather than the vehicle’s appearance. Second, for a forward-facing camera it is only *partially visible*: at any instant the camera sees a limited stretch of road, and quantities such as the curvature of the lane the vehicle is about to enter are not fully determined by the current frame. A single-frame encoder is therefore insufficient; the road context must be aggregated over time.

Temporal road-context encoder... We condition the physical encoder on a history of visual latents:

$$(z_t^v, z_t^r) = \mathcal{E}_p(z_{t-k+1:t}), \quad z_\tau = \mathcal{E}_v(y_\tau), \quad (10)$$

where k is the context-window length. The Transformer-based physical encoder handles variable-length temporal inputs via self-attention, allowing it to aggregate motion and appearance cues across frames: as the vehicle advances, the road geometry that was previously seen far ahead becomes the geometry the vehicle is on, so a temporal window disambiguates cross-track error, heading error, and curvature that a single frame leaves ambiguous. The encoder produces both components jointly but supervises them separately through (9), ensuring z_t^r carries genuine environmental information rather than being absorbed into the vehicle component.

Privileged supervision for real-world deployment... On the real DonkeyCar, the road context cannot be measured by the deployed car itself. We therefore use *privileged supervision*: during training, an external localization system (a known track map together with the logged car pose, as is also obtainable from an overhead camera) provides ground-truth road context Ξ^r for each frame; at test time this signal is withheld and the deployed car observes only its front camera. The encoder (10) is thus trained to predict, from the onboard image stream alone, the road context that the privileged signal supplied during training. This is precisely a teacher–student / learning-from-privileged-information setup [76]: the privileged road context teaches the encoder what to extract, and at deployment the encoder reconstructs it from images. Because z^r is physically interpretable, the recovered road context can be inspected and compared against the true lane geometry, which we use as a diagnostic in Section 7.

5.3. Predicting the Road Environment

Challenge and motivation... Encoding the road at the current step is necessary but not sufficient. To roll out a trajectory over a long horizon without new observations, the world model must anticipate how the road context itself changes as the vehicle advances: the curve that is ahead now is underfoot a few steps later, and the cross-track and heading errors evolve as the vehicle tracks (or departs from) the lane. A model that predicts only the vehicle state, holding the road context fixed or ignoring it, drifts quickly because it steers the predicted vehicle through a road that no longer matches reality.

We therefore make road prediction an explicit, physically interpretable part of the dynamics.

Split dynamics.. We use a *split-dynamics* model that advances the vehicle and the road context with separate, physically appropriate transitions:

$$\hat{z}_{t+1}^v = \phi_\theta(z_t^v, a_t), \quad (11)$$

$$\hat{z}_{t+1}^r = f_r(z_t^r, z_t^v, a_t; \theta_r). \quad (12)$$

The vehicle transition ϕ_θ is the learnable bicycle kinematic model (below); the road transition f_r is a structured, geometry-driven model (below). Splitting the dynamics decouples the fast-varying vehicle motion from the slow-varying road geometry and lets each receive its correct inductive bias.

Bicycle kinematic model for the vehicle.. For driving environments we replace the generic second-order dynamics of the conference paper with a *bicycle kinematic model*, a physically appropriate prior for wheeled-vehicle motion:

$$x_{t+1} = x_t + v_t \cos(\psi_t + \beta_t) \Delta t, \quad (13)$$

$$y_{t+1} = y_t + v_t \sin(\psi_t + \beta_t) \Delta t, \quad (14)$$

$$\psi_{t+1} = \psi_t + \frac{v_t}{l_r} \sin(\beta_t) \Delta t, \quad (15)$$

$$v_{t+1} = v_t + a_t \Delta t, \quad (16)$$

where $\beta_t = \arctan\left(\frac{l_r}{l_f + l_r} \tan(\delta_{f,t})\right)$ is the slip angle, l_f and l_r are the distances from the center of mass to the front and rear axles, $\delta_{f,t}$ is the steering angle, v_t is the longitudinal speed, and ψ_t is the heading. The wheelbase parameters l_f, l_r are unknown and learned during training. The vehicle-relative dynamics (11) apply (13)–(15) in the local frame of each window, expressing displacements relative to the initial timestep (3).

Structured road-context transition.. The road-context state $(e_{\text{CTE}}, e_\psi, \kappa)$ evolves slowly and geometrically. As the vehicle moves by the body-frame increment implied by its predicted motion, the road ahead is effectively re-indexed: the lane geometry is advected past the vehicle. We therefore model f_r as a geometry-driven transition that (i) updates the cross-track error and heading error from the vehicle’s predicted lateral and angular motion relative to the road, and (ii) advances the curvature conservatively, treating it as locally

constant over a single step and updating it toward the curvature of the next road segment:

$$z_{t+1}^r = f_r(z_t^r, z_t^v, a_t; \theta_r). \quad (17)$$

A small learnable residual corrects for effects not captured by the rigid geometric update (e.g., imperfect lane tracking), with θ_r learned jointly. Crucially, because f_r operates on the physically interpretable z^r , the forecasted road context at any horizon is itself interpretable: the model outputs a predicted lane geometry that can be visualized and checked, not an opaque feature.

5.4. Training Procedure

Training follows a staged pipeline. Stage 1 trains the vision autoencoder $(\mathcal{E}_v, \mathcal{D}_v)$ on image reconstruction. Stage 2 trains the physical encoder $\mathcal{E}_p$ to produce the factorized interpretable state, optimizing (9) jointly with a latent reconstruction term. Stage 3 trains the split-dynamics models (ϕ_θ for the vehicle and f_r for the road context), with an optional end-to-end fine-tuning pass that updates the encoder and dynamics together to align the encoded state with what the dynamics can predict. Algorithm 1 summarizes the procedure.

6. Experimental Evaluation

6.1. Environments and Evaluation Tiers

We evaluate across three tiers of increasing difficulty. Tier 1 establishes continuity with the conference paper; Tiers 2 and 3 are the two new partially observable driving case studies (Contribution 1).

Tier 1: Fully observable benchmarks.. **CartPole** and **Lunar Lander** [81] are retained from the conference paper as controlled settings where the system state is a complete Markov description. These results establish continuity with the conference version and a clean baseline for physically interpretable prediction under simple dynamics.

Tier 2: Partially observable simulated driving (CarRacing).. **CarRacing** [81] is a top-down simulated racing environment in which the vehicle navigates a procedurally generated track. The observation is a bird’s-eye-view RGB image (native 96×96 , resized to 64×64 for the encoder). The vehicle’s kinematic state is available from the simulator, but the road geometry (the

Algorithm 1 PIWM training for partially observable driving.

Input: Training set $\mathcal{S}$, loss weights $\lambda_v, \lambda_r, \lambda_{\text{dyn}}$, context length k

- 1: Train vision autoencoder $(\mathcal{E}_v, \mathcal{D}_v)$ on image reconstruction
 - 2: **for** each batch $(y_{1:M}, a_{1:M}, \Xi_{1:M}^v, \Xi_{1:M}^r)$ **do**
 - 3: $z_t \leftarrow \mathcal{E}_v(y_t)$ for all t
 - 4: $(z_t^v, z_t^r) \leftarrow \mathcal{E}_p(z_{t-k+1:t})$ $\triangleright$ encode vehicle and road (Sec. 5.2)
 - 5: Update $\mathcal{E}_p$ with $\lambda_v \mathcal{L}_{\text{interp}}(z_t^v, \Xi_t^v) + \lambda_r \mathcal{L}_{\text{interp}}(z_t^r, \Xi_t^r) + \lambda_{\text{latent}} \mathcal{L}_{\text{recon}}$
 - 6: **end for**
 - 7: **for** each batch **do**
 - 8: Obtain (z_t^v, z_t^r) from the encoders
 - 9: $\hat{z}_{t+1}^v \leftarrow \phi_\theta(z_t^v, a_t)$ $\triangleright$ bicycle model
 - 10: $\hat{z}_{t+1}^r \leftarrow f_r(z_t^r, z_t^v, a_t; \theta_r)$ $\triangleright$ road prediction (Sec. 5.3)
 - 11: Update θ (bicycle) and θ_r (road) on the multi-step rollout loss $\|\hat{z}_{t+1}^v - \hat{\mu}_{\Xi_{t+1}^v}^v\|^2 + \|\hat{z}_{t+1}^r - \hat{\mu}_{\Xi_{t+1}^r}^r\|^2$
 - 12: **end for**
 - 13: (Optional) End-to-end fine-tune $\mathcal{E}_p, \phi_\theta, f_r$ jointly
 - 14: **return** trained model parameters
-

shape of the upcoming track and the vehicle’s relationship to the lane) is not part of the state and must be inferred from the image. Because the camera is top-down, the road ahead is visible but the model must still *encode* it into the physical road context and *predict* its evolution during rollout. This is the primary controlled testbed for the factorized-state formulation and split dynamics.

Tier 3: Real-world DonkeyCar deployment.. The **DonkeyCar (real)** tier evaluates the factorized formulation on a physical RC-scale autonomous car driven on a laboratory track. The car observes its environment through a front-facing camera (the test-time observation, 64×64), and logs wheel-encoder speed, steering, and pose. This is the most demanding tier: the camera is forward-facing, so the road ahead is only partially visible, and the road context cannot be measured by the deployed car. We collect driving trajectories on the physical platform and derive ground-truth road context from external track localization, used as privileged supervision at training time only (Section 5.2); at test time the model infers and rolls out road context from the onboard front-camera stream alone.

6.2. Experimental Setup

Benchmarks (Tier 1)... The data-collection and weak-supervision protocol follows the conference paper. For each environment, we collect 60,000 trajectories of at least 50 steps, generated by a mix of random actions and a pretrained neural controller. Weak-supervision labels are synthesized by adding biased uniform noise of relative width $\delta \in \{0\%, 5\%, 10\%\}$ to the ground-truth state, as described in Appendix Appendix A.

CarRacing (Tier 2)... Driving data are generated with a pretrained controller on procedurally generated tracks; each episode records the onboard image, action, and simulator state. Ground-truth labels for the vehicle-relative state $(x_{\text{rel}}, y_{\text{rel}}, \psi_{\text{rel}})$ and the road-context state $(e_{\text{CTE}}, e_{\psi}, \kappa)$ are available directly from the simulator and its track geometry. We use exact ground-truth labels for training (corresponding to $\delta = 0$) and evaluate whether the factorized-state formulation improves long-horizon prediction relative to baselines that use only vehicle-state labels.

DonkeyCar real (Tier 3)... We drive the physical DonkeyCar on a closed laboratory track and record synchronized front-camera frames (64×64), control actions (steering, throttle), and vehicle pose, at approximately 22 fps. Trajectories are segmented into clean contiguous laps; brief sensing artifacts (dropped or interpolated frames, pose glitches) are filtered out before training. Ground-truth road context is computed from the known track centerline and the logged pose (privileged supervision), and is withheld at test time. We hold out a subset of laps for evaluation and report prediction error on the held-out laps.

Architecture and training... The driving experiments use the extrinsic PIWM variant (a vision encoder followed by a Transformer physical encoder producing the factorized state), which the conference paper showed to be the strongest configuration. The vision encoder maps each 64×64 observation to a compact latent. The physical encoder uses a temporal context window to infer road context from image sequences (Section 5.2). Dynamics are trained with a cosine learning-rate schedule and early stopping on validation loss. All baselines are trained under identical conditions with the same data, optimizer (Adam), and evaluation protocol.

6.3. Baselines

We compare against the same families used in the conference paper, augmented with ablations that isolate the road-context contributions. Data-driven sequence models (**LSTM** and **Transformer**) serve as non-physical benchmarks. **DVBF** [45], **GokuNet** [43], and **Vid2Param** [59] are physics-aware baselines, and **SindyC** [60] is a classical dynamics-discovery baseline operating on the interpretable latent. For driving, to isolate the contribution of road-context encoding and prediction, we additionally include two PIWM ablations: **PIWM-linear** (PIWM with a linear vehicle-dynamics model and no road context) and **PIWM-bicycle** (the bicycle model on the vehicle state, again without the road-context factorization). The gap between these ablations and the full model measures the value of encoding and predicting the road, not just the vehicle.

6.4. Metrics

The primary metric is the root mean squared error (RMSE) of the predicted state over multi-step rollout horizons. For benchmark tasks, RMSE is computed over all state dimensions. For driving tasks, RMSE is evaluated on the vehicle-relative state:

$$\text{RMSE}_{xy} = \sqrt{\frac{1}{N} \sum_{i=1}^N ((\hat{x}_{\text{rel},i} - x_{\text{rel},i})^2 + (\hat{y}_{\text{rel},i} - y_{\text{rel},i})^2)}. \quad (18)$$

We additionally report the road-context prediction error (on $e_{\text{CTE}}, e_{\psi}, \kappa$) to evaluate the quality of the predicted road environment directly. We evaluate at horizons of 30 steps (benchmarks) and up to 100 steps (driving). For environments where ground-truth physical parameters are known (CartPole, Lunar Lander), we additionally report relative parameter-recovery error.

7. Results

7.1. Benchmark Results (Tier 1)

The fully observable benchmark results are carried over from the conference paper [8] and summarized here for completeness. Across CartPole, Lunar Lander, and DonkeyCar, the extrinsic-discrete PIWM variant consistently achieves the lowest and most stable 30-step prediction error, substantially outperforming data-driven and physics-aware baselines; the quantized

Table 1: CarRacing longitudinal-displacement (x_{rel}) prediction error at increasing horizons (lower is better). PIWM with the bicycle model and road-context factorization maintains low error to 100 steps, whereas the vehicle-only ablations (PIWM-linear, PIWM-bicycle) and most baselines diverge. *The numerical entries are populated from our evaluation runs; confirm against the final experiment freeze before submission.*

Model	step 20	step 50	step 75	step 100
PIWM-linear	0.015	0.124	0.559	1.985
PIWM-bicycle (veh. only)	0.008	0.186	0.696	2.020
SindyC	0.046	0.927	4.493	15.03
DVBF	0.009	0.182	1.242	3.556
Vid2Param	0.001	0.094	1.093	6.044
GokuNet	0.001	0.038	0.324	1.405
PIWM (ours, full)	0.002	0.026	0.074	0.306

latent provides strong regularization that improves long-horizon stability. PIWM also recovers the ground-truth physical parameters of CartPole and Lunar Lander with low relative error across all noise levels δ , confirming that the learned representations are genuinely physically grounded rather than merely correlated with the physics. These benchmark results establish that the driving extension is a proper superset of the original framework: on fully observable tasks, where the vehicle state is already Markov, PIWM behaves exactly as before.

7.2. CarRacing Results (Tier 2)

Figure 5 reports state error over a 100-step prediction horizon on CarRacing. Table 1 summarizes the longitudinal-displacement (x_{rel}) error at representative horizons.

The proposed PIWM with the bicycle model and factorized state substantially outperforms all baselines across the state dimensions and across the full 100-step horizon. Critically, the two ablations that drop the road context, PIWM-linear and PIWM-bicycle (vehicle state only), are competitive at short horizons but diverge as the horizon grows: by 75–100 steps their error is roughly an order of magnitude larger than the full model. This is the central empirical signature of partial observability. A vehicle-only model cannot anticipate the upcoming road geometry, so once the rollout enters curvature that was not represented, errors compound; the full model, which encodes the road context and predicts its evolution, steers the predicted ve-

hicle through the correct geometry and stays accurate. Data-driven baselines (LSTM-style and GokuNet) and the physics-aware DVBF and Vid2Param exhibit rapid error growth at long horizons, consistent with the conference findings.

Quality of the predicted road environment.. Because z^r is interpretable, we can inspect the forecasted road directly. The predicted road context (cross-track error, heading error, curvature) tracks the ground-truth lane geometry over the rollout, confirming that the model is not merely improving vehicle prediction as a side effect but is genuinely forecasting the external environment in physically meaningful terms.

7.3. Real DonkeyCar Results (Tier 3)

We deploy the learned model on the physical DonkeyCar, where the road context is supervised only during training and must be inferred from the front camera at test time.

Short-horizon accuracy.. For one-step prediction from a temporal observation context, the model attains a relative displacement error of $\text{RMSE}_x = 0.057\text{ m}$, $\text{RMSE}_y = 0.048\text{ m}$, and $\text{RMSE}_{xy} = 0.075\text{ m}$, i.e. an average positional error of roughly 7.5 cm on the physical car.

Per-lap analysis.. Prediction error varies across laps. Most laps yield consistent, low error; a few exhibit larger error traceable to sensing artifacts in the recorded trajectory (occasional pose irregularities from logging noise) rather than model failure. To characterize the model under clean conditions, we report a best-lap analysis: on the best-performing lap the model attains

$$\text{RMSE}_{xy} = 0.0238\text{ m},$$

roughly 2.4 cm of positional error. Figure 6 compares the predicted trajectory against the ground-truth path for this lap; the predicted displacements closely follow the true trajectory, demonstrating accurate motion prediction on real hardware when the underlying data are clean.

Long-horizon rollout.. We additionally evaluate multi-step (autoregressive) rollout over complete laps, feeding the predicted state and the control inputs back into the split-dynamics model. Across laps, the long-horizon RMSE_{xy} remains in the range 0.05 m–0.07 m, indicating that error stays bounded as

the horizon grows. Figure 7 shows representative long-horizon predictions and the predicted lane overlaid on the ground-truth track. The trajectories remain well aligned over most of the lap, with the largest deviations appearing in high-curvature segments, where small per-step errors accumulate; nonetheless the predicted road context keeps the rollout physically plausible. This is the key real-world finding: a factorized representation whose road context is learned under privileged supervision transfers to deployment, where the road must be recovered from the onboard camera alone.

8. Discussion

Encoding and predicting the road is what restores long-horizon accuracy.. The CarRacing results show a clear and growing gap between models that include the road-context factorization and those that do not: vehicle-only ablations match the full model at short horizons but diverge by 50–100 steps (Table 1). This is the empirical content of the partial observability argument from Section 4. Vehicle state alone is non-Markov; the road context carries the missing information about upcoming geometry, and both *encoding* it from images and *predicting* its evolution are required for stable rollout.

Bicycle dynamics vs. generic dynamics.. The advantage of the bicycle model over a generic linear predictor indicates that the physical inductive bias is beneficial beyond raw capacity: the bicycle model constrains predicted motion to physically plausible, heading-consistent trajectories, which reduces error accumulation over long horizons. This parallels the conference finding for benchmark tasks, where structured dynamics outperformed black-box sequence models.

Interpretability of the forecasted environment.. Because the road context is physically interpretable, the forecasted road, not just the forecasted vehicle, can be inspected and monitored. On both CarRacing and the real DonkeyCar, the predicted lane geometry can be compared against ground truth, providing a diagnostic that opaque world models do not offer. This is directly useful for safety monitoring: a forecasted road that departs from plausible geometry is a detectable failure signal.

Privileged training for real deployment.. The real DonkeyCar uses privileged road-context supervision at training and withholds it at test. This reflects practical deployability: external localization (a mapped track, or an overhead

view) is available in instrumented training environments but not on a freely deployed car. The model must therefore learn to read road context from onboard images, guided during training by the privileged signal. The fact that the learned encoding transfers to the unaided setting is what makes the approach viable on real hardware.

Limitations.. First, the road-context variables are specified manually for each environment; a more general approach might discover the decomposition automatically. Second, the bicycle model is a kinematic approximation that ignores tire dynamics and slip, which may limit accuracy at high speeds or on low-friction surfaces. Third, privileged supervision requires an instrumented training environment to obtain ground-truth road context; reducing this requirement (e.g., via self-supervised road estimation) is open. Finally, the method currently assumes a single vehicle and a single lane; multi-agent and multi-lane settings are future work.

Future directions.. Natural next steps include scaling PIWM to open-world driving with richer road topology, multiple agents, and additional sensor modalities (LiDAR, radar); treating other vehicles as additional interpretable road-context variables to handle interactions and occlusions; and discovering the road-context structure from data via symbolic regression or causal structure learning, which would reduce the need for environment-specific design.

9. Conclusion

This article extended the Physically Interpretable World Model (PIWM) from fully observable benchmarks to partially observable driving. The central observation is that driving is partially observable: the vehicle’s physical state alone is non-Markov with respect to future trajectory, because the road context (lane geometry and the vehicle’s relationship to it) is absent from any standard state representation and only partially visible to an on-board camera. We addressed this along three axes. We introduced two partially observable driving case studies, a simulated CarRacing track and a real physical DonkeyCar. We introduced a physically interpretable *encoding* of the road environment, factorizing the interpretable state into a vehicle-relative component and a road-context component inferred from images, including under privileged supervision for real-world deployment. And we introduced a physically interpretable *prediction* of the road environment,

a split-dynamics model that forecasts how the road context evolves alongside a learnable bicycle model for the vehicle. Experiments on CarRacing and the real DonkeyCar confirm that encoding and predicting the external environment, not merely the vehicle, is what enables stable long-horizon prediction under partial observability, while the original benchmark results are retained unchanged. Physically interpretable world modeling for driving thus requires aligning latent representations with physical variables *and* decomposing those representations according to the semantic structure of the task, so that the external environment is both encoded and predicted in physically meaningful terms.

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Appendix A. Weak Supervision Noise Model

The weak supervision in Tier 1 benchmark experiments follows the noise model from the conference paper. For the i -th state component with valid range size $\mathcal{X}_i$, we generate a randomly shifted center $\tilde{\mu}_i = x_i + \Delta_i$, $\Delta_i \sim \text{Unif}[-\frac{1}{2}\delta|\mathcal{X}_i|, \frac{1}{2}\delta|\mathcal{X}_i|]$, and draw the supervisory distribution as $p_i(x) = \text{Unif}[\tilde{\mu}_i - \frac{1}{2}\delta|\mathcal{X}_i|, \tilde{\mu}_i + \frac{1}{2}\delta|\mathcal{X}_i|]$. At each timestep, $L = 50$ samples are drawn to form the proxy supervision set Ξ_t . For Tier 2 and Tier 3 experiments, exact ground-truth labels are used at training time ($\delta = 0$), reflecting the availability of simulator state and training-time track localization; at test time, the road context is withheld and inferred from images.

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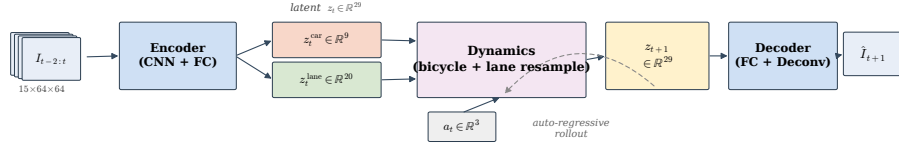
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Figure 1: Partial observability in driving. Two vehicles with identical kinematic states (x, y, ψ, v) diverge because the road geometry ahead differs. The vehicle state alone is non-Markov: the road context r_t (lane shape and the vehicle's relation to it) is missing and, for a forward-facing camera, only partially visible.

(a) PIWM inference pipeline



(b) Inside the dynamics block — analytic dynamic-bicycle for the car branch, resample-based propagation for the lane branch

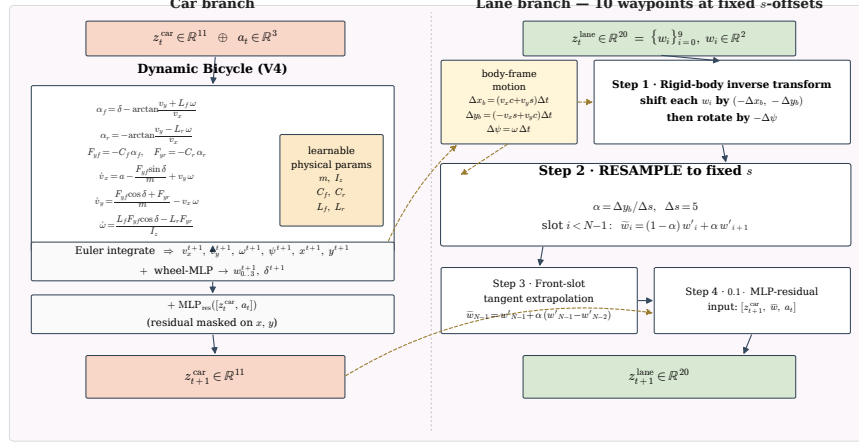


Figure 2: Overview of the PIWM architecture. A vision encoder maps observations to a visual latent; a physical encoder extracts the factorized interpretable state $z^* = [z^v, z^r]$ (vehicle-relative state and road context); and a split-dynamics model rolls the state forward, applying a learnable bicycle kinematic model to z^v and a structured, geometry-driven transition to z^r . The encoder (Section 5.2) infers the external road environment from images; the predictor (Section 5.3) forecasts how that environment evolves.

Lane waypoint extraction — built once per episode by `scripts/lane\_preprocess.py`

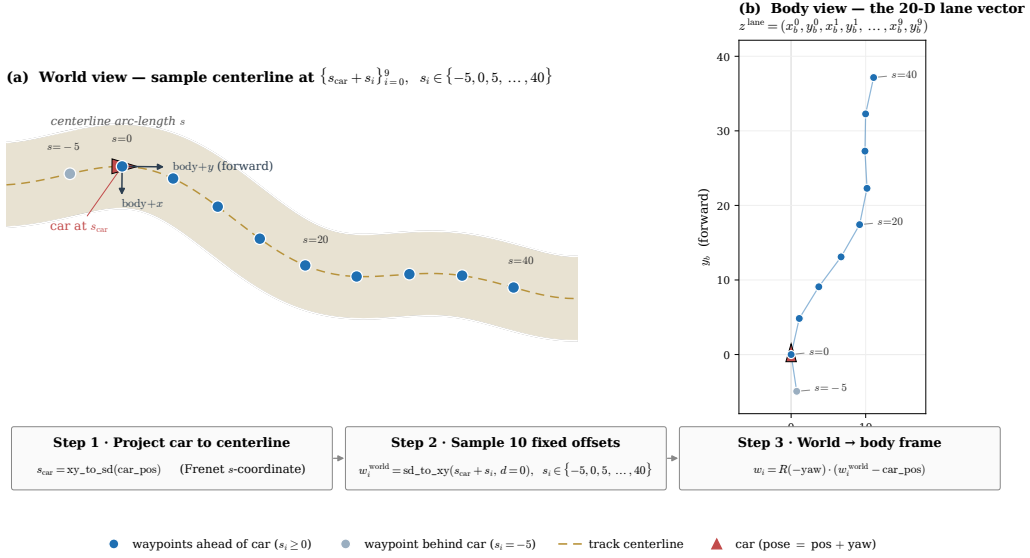


Figure 3: Encoding the road environment. The road-context state z^r (cross-track error e_{CTE} , heading error e_ψ , curvature κ) describes the lane ahead in the vehicle’s body frame and is inferred from a temporal window of onboard images. On the real DonkeyCar this encoding is supervised by privileged track localization at training time and recovered from the front camera alone at test time.

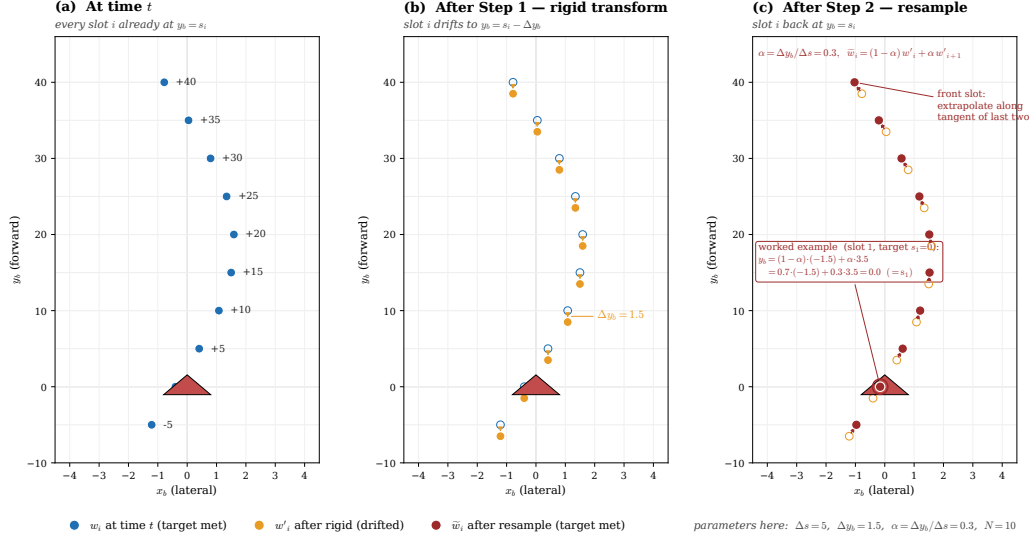


Figure 4: Predicting the road environment. As the vehicle advances, the road context is advected past it: the geometry ahead is re-indexed under the predicted ego motion (a quasi-rigid geometric update) and corrected by a small learned residual. Because the predicted road context remains physically interpretable, the forecasted lane can be inspected at every horizon.

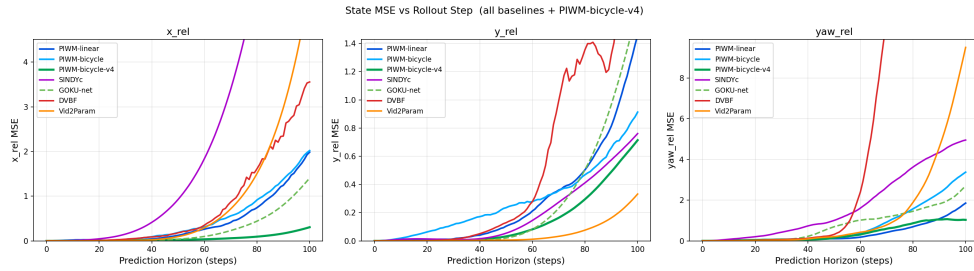


Figure 5: Prediction performance on CarRacing. State error over a 100-step horizon for x_{rel} , y_{rel} , and ψ_{rel} , comparing PIWM (with factorized state and split dynamics) against PIWM-linear, PIWM-bicycle (vehicle state only, no road context), SindyC, GokuNet, DVBF, and Vid2Param. Methods that encode and predict the road context remain accurate at long horizons, while vehicle-only and data-driven baselines diverge.

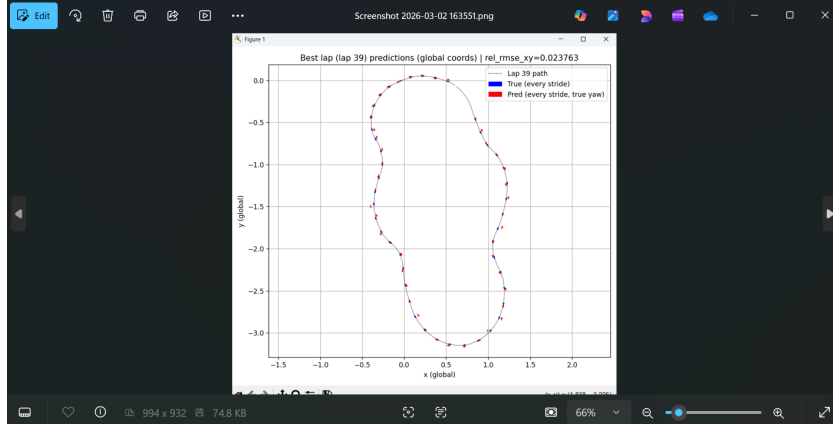


Figure 6: Real DonkeyCar best-lap prediction. Predicted trajectory (in global coordinates) compared with the ground-truth path; the relative displacement error is $\text{RMSE}_{xy} = 0.0238 \text{ m}$ ($\approx 2.4 \text{ cm}$).

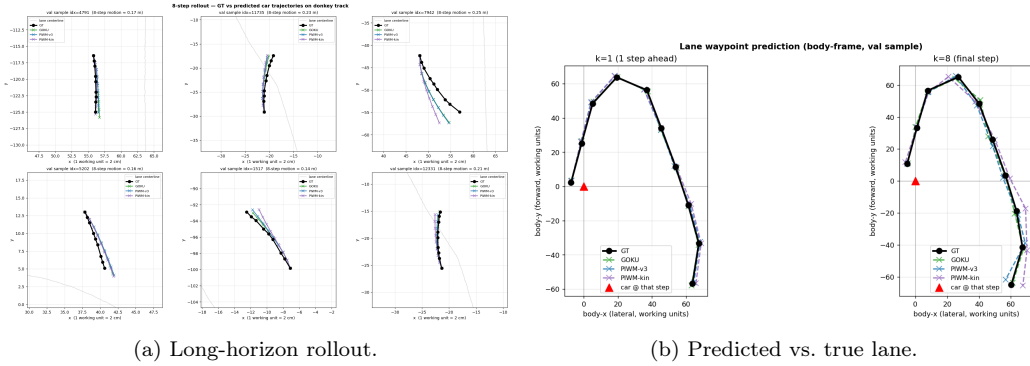


Figure 7: Real DonkeyCar long-horizon behavior. (a) Predicted vs. ground-truth trajectory over a lap. (b) The predicted road context (lane geometry) overlaid on the true lane, showing that the model forecasts the external environment, not only the vehicle.